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Basic Configuration for RHEL & Rocky 9.x (v.2024)

APPROVED STATE

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first frame networkers

Document information / intro

This document describes the basic configuration for RedHat Enterprise Linux (RHEL) 9.x.

About linux at first frame

- RHEL is the first frame prefered linux distribution.
- Licensing:
  - first frame preferes CSP licensing for RHEL. Use of other license models needs to be approved by CCL APD. (Hint: first frame needs CSP functionality for delivery of managed services, ex. RedHat cloud console)
  - Developer subscriptions are not intended for production workloads
- Commercial/Enterprise grade vendor support is available from RedHat.
- Only RHEL is supported for feature complete managed server linux according to service catalog.

References

add references here

Configuration

1	**Source** • RHEL: https://access.redhat.com/downloads/content/rhel (use rhel 9.4 boot.iso) • Rocky: https://rockylinux.org/download (use boot.iso)
2	**Hardware** • Reference is optimized for apply to virtual machines • CPU: 2 vCPU • Memory: 4 GB • Disk: 32 GB

3	**Anaconda installer** • Language: English (US) • Keyboard: Swiss German • Software: Minimal Install + Standard Software • Network: • Hostname: enter unique hostname • IP/Netmask: enter static ip address • Gateway: enter default gateway • DNS: use customer dns server if exists, else use quad9 quad9: 9.9.9.9, 149.112.112.112 • Kdump: disabled • User: enable Option "Make this user administrator" for admin user created by anaconda installer; prevent using root user for system administration later • Partitions • /, 8 GiB, ext4 • /boot, use predefined size, ext4 • /var, 8GiB, ext4 • RHEL only • use customer specific activation key https://console.redhat.com/insights/connector/activation-keys • organization: 15108280 • activation key: <customer id>-rhel-<8 lcase secret, first 8 char of new guid> • disable rhel insights (configure later)

4	**Hostname (configuring hostname is required if the installation is built from kickstart template)**
	`# show current hostname`
	`hostnamectl status`
	
	`# configure hostname`
	`sudo hostnamectl set-hostname testserver1.domain.tld`

5	**Network settings** `sudo nmtui`

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Subscription manager and insights client (rhel only)

```
# configure tag
echo "group: '<customer id>'" | sudo tee /etc/insights-client/tags.yaml

# register subscription manager
subscription-manager register --org=<org id> --activationkey=<key>

# register insights client
sudo insights-client --register

# restart insights client
sudo systemctl restart insights-client

# get status of insights client
sudo systemctl status insights-client
```

Note: reset/cleanup subscription manager (rhel only, only required for troubleshooting reasons)

```
# unregister
sudo subscription-manager unregister

# reset
sudo subscription-manager clean

# should already by reseted by clean command
sudo rm -f /etc/pki/consumer/*
sudo rm -f /etc/pki/entitlement/*

# remove system from inventory
# https://console.redhat.com/insights/inventory
```

7

Remove software components

```
# uninstall cockpit
sudo dnf remove -y cockpit*
sudo firewall-cmd --remove-service=cockpit --permanent
```

8

Update Software

```
sudo dnf update
```

9

Configure boot splash screen

```
# boot splash screen (show config)
sudo grubby --info=ALL

# boot splash screen (edit config)
sudo grubby --update-kernel=ALL --remove-args="rhgb quiet"
```

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Configure power profile

```
# power profile (show config)
sudo tuned-adm active

# power profile (edit config)
sudo tuned-adm profile virtual-guest
```

11

Enable persistent logging

```
# enable persistent logging
sudo mkdir -p /var/log/journal
sudo systemctl --signal=USR1 kill systemd-journald.service
```

12

Configure SSHd

```
# configure sshd
sudo vim /etc/ssh/sshd_config

---
PermitRootLogin no
ClientAliveInterval 300
ClientAliveCountMax 3
UseDNS no
---

# restart sshd
sudo systemctl restart sshd.service
```

13

Configure sudo

```
# disable pw prompt for sudo command
sudo visudo

---
## Allows people in group wheel to run all commands
# %wheel        ALL=(ALL)       ALL

## Same thing without a password
%wheel  ALL=(ALL)       NOPASSWD: ALL
---
```

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Set password

```
# change root password
sudo passwd root

# change 'tux' password
passwd tux
```

15

Create login/password for 3rd party (ex. supplier for application)

```
# create user
sudo useradd <username>

# add user to wheel (sudo) group
sudo usermod -aG wheel <username>

# set password
sudo passwd <username>
```

Remote Support Systems

Monitoring

1

configure SNMP

```
# install snmp
sudo dnf install net-snmp

# enable snmp service
sudo systemctl enable snmpd.service

# create user on /var/lib/net-snmp/snmpd.conf
# authorize user on /etc/snmp/snmpd.conf
sudo net-snmp-create-v3-user -ro -A <auth pass> -X <priv pass> -a SHA -x AES snmpuser-<random 8 char>

# configure firewall (filter by source!!!)
sudo firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="<probe ip>/32" port
protocol="udp" port="161" accept'
sudo firewall-cmd --reload
```

disable snmp v1/v2: disable line on /etc/snmp/snmpd.conf

```
#com2sec notConfigUser  default       public
```

disable rouser default access: disable line on /etc/snmp/snmpd.conf

```
#rouser snmpuser-3d7c97bc
```

add line to /etc/snmp/snmpd.conf

```
# views hostResourcesMibModule (common mib for os monitoring)
# hrSystem
view custom_view included .1.3.6.1.2.1.25.1
# hrStorage
view custom_view included .1.3.6.1.2.1.25.2
# hrDevice
view custom_view included .1.3.6.1.2.1.25.3

# groups
group custom_group usm snmpuser-<random 8 char>

# access
access custom_group "" any authPriv exact custom_view none none
```

2	**PRTG Configuration** • Configure 'Credentials for SNMP Devices' on group layer (ex. Devices > Probe Layer > Compute)
3	**Base operating system sensors** • Ping • Uptime • CPU • Physical Memory • Swap • Disks (at least /, /boot, /var) • Error threshold < 20%

RedHat insights

1	see "Configuration > Subscription manager and insights client (rhel only)"

RMM

1	Definition von Vorgaben in diesem Dokument nicht erforderlich. RMM wird für Linux zum Zeitpunkt der Freigabe dieser Referenzkonfiguration nicht genutzt.

MCv2

1	Definition von Vorgaben in diesem Dokument nicht erforderlich.

Test

1	subscription manager is activated (RHEL only)
2	all software updates are installed
3	system is registered on RedHat insights and is sending data (RHEL only)
4	LVM partition is the last partiton on the disk - it must be possible to extend lvm partition without moving other partitions
5	ext4 is used as default file system - only use xfs if required by application (ex. veeam backup repository)